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Learning scientist, mathematics educator, and Balinese dance practitioner investigating how culturally situated movement practices can become resources for mathematical learning. Drawing on embodied cognition, design-based research, and multimodal interaction analysis, my work examines how bodily perception, gesture, spatial coordination, and cultural knowledge support reasoning and inform educational design.

EDUCATION

2026 May	Ph.D. in Education Learning Science and Human Development University of California, Berkeley, United States Thesis: *Grounding geometry in culturally authentic movement practices: From attentional anchors to auxiliary lines in Balinese Dance* (supervisor: Professor Dor Abrahamson)
2016 March	M.A. in Mathematics Education (with distinction) Universitas Pendidikan Ganesha, Indonesia Thesis: *Teaching addition and subtraction using an empty number line to enhance second-grade students' ability in number sense*
2015 July	M.Sc. in Mathematics Education (with distinction) Utrecht University, The Netherlands & Universitas Sriwijaya, Indonesia Thesis: *Bridging between arithmetic and algebra: Using patterns to promote algebraic thinking*
2012 June	B.A. in Mathematics Education Universitas Pendidikan Ganesha, Indonesia Thesis: *The impact of IKRAR learning model with local wisdom orientation on the students' problem-solving ability*

SELECTED DANCE PERFORMANCE

Since 2021, I have performed Balinese dance at more than thirty public events across Northern California, primarily with Gamelan Sekar Jaya, a Bay Area organization dedicated to Balinese gamelan and dance. Several of these performances, marked with an asterisk (*), were presented at the invitation of the Consulate General of the Republic of Indonesia in San Francisco.

Nov 2021	Taste of Indonesia (culinary and dance & gamelan performance), Richmond, CA, with Gamelan Sekar Jaya. Dance performed: Rejang Penganter Alit.*

Mar 2022	Open House, Hari Saraswati celebration, Berkeley, CA, with Gamelan Sekar Jaya. Dances performed: Rejang Sudamala; Condong.
Jun 2022	End-of-year performance, Berkeley, CA, with Gamelan Sekar Jaya. Dance performed: Rejang Sudamala.
Aug 2022	Summer at the Hidden Cafe, Berkeley, CA, with Gamelan Sekar Jaya. Dance performed: Rejang Sudamala.
Aug 2022	Bazaar Nusantara 2022, Sacramento, CA, organized by Perhimpunan Masyarakat Indonesia (Indonesian Community Association). Dances performed: Condong; Joged Bumbung.*
Sep 2022	Solano Stroll Festival, Solano, CA, with Gamelan Sekar Jaya. Dance performed: Joged Jegog.
Sep 2022	Reception at the Consulate General of the Republic of Indonesia, San Francisco, CA, with Gamelan Sekar Swarasanti. Dance performed: Joged Jegog.*
Sep 2022	North Columbia Schoolhouse Cultural Center, Nevada City, CA, with Gamelan Sekar Swarasanti. Dance performed: Joged Jegog.
Sep 2022	Bay Light Mixer Art (Cultural, Tourism, and Economic Promotion Program), San Francisco, CA, organized by the Commonwealth Club of California; invited as a representative of the Consulate General of the Republic of Indonesia. Dance performed: Condong.*
Nov 2022	Open House, Hari Saraswati celebration, Berkeley, CA, with Gamelan Sekar Jaya. Dances performed: Rejang Sudamala; Panyembrahma.
Nov 2022	Ashkenaz show, Berkeley, CA, with Gamelan Sekar Swarasanti. Dance performed: Joged Jegog.
Nov 2022	Oakland Asian Cultural Center exhibition, Oakland, CA, with Gamelan Sekar Jaya. Dance performed: Pendet.
Mar 2023	Concert for the elderly community, Oakmont, CA, with Gamelan Sekar Jaya. Dance performed: Joged Jegog.
May 2023	Asian American Pacific Islander Heritage Month reception and performance, San Francisco, CA; representative of the Consulate General of the Republic of Indonesia. Dance performed: Panyembrahma.*
May 2023	Asian American Native Hawaiian and Pacific Islander Heritage Month Festival, San Francisco, CA. Dance performed: Candra Metu.
Jun 2023	Craft workshop (Balinese offering-making) and performance, Rhythmix, Alameda, CA, with Gamelan Sekar Jaya. Dance performed: Joged Jegog.
Jul 2023	Educational event, Claremont Library, Berkeley, CA, with Gamelan Sekar Jaya. Dance performed: Panyembrahma.
Aug 2023	49th Annual Nihonmachi Street Fair, San Francisco, CA. Dances performed: Puspanjali; Legong Keraton Lasem.
Feb 2024	Celebration of Indonesian Diaspora, International House, UC Davis, Davis, CA. Dance performed: Pendet.
Dec 2024	Holiday Craft Fair, Studio One Art Center, Oakland, CA, with Gamelan Sekar Jaya. Dances performed: Pendet Angklung; Joged Jegog.

Dec 2024	Night of Balinese Dance, Ashkenaz Music and Dance Community Center, Berkeley, CA, with Gamelan Sekar Jaya. Dance performed: Sekar Harum.
Feb 2025	Open House, Berkeley, CA, with Gamelan Sekar Jaya. Dance performed: Pendet Penyambutan.
Apr 2025	Indonesia & India in California Festival, Berkeley, CA, with Gamelan Sekar Jaya. Dance performed: Pendet Peliatan.
Apr 2025	4th Asian Cultural Festival, Berkeley, CA, with Gamelan Sekar Jaya. Dance performed: Pendet Angklung.
Apr 2025	UC Berkeley Noon Concert, Berkeley, CA, with the UC Berkeley Gamelan class and Gamelan Sekar Jaya. Dance performed: Gegaboran.
May 2025	Friends of Indonesia 4th Annual Bazaar, San Francisco, CA. Dances performed: Pendet Peliatan; Legong Kuntir.*
Oct 2025	Open House, Berkeley, CA, with Gamelan Sekar Jaya. Dance performed: Pendet Angklung.
Nov 2025	Art gallery opening, San Francisco, CA. Dance performed: Legong Kuntir.
Dec 2025	Night of Balinese Dance, Ashkenaz Music and Dance Community Center, Berkeley, CA, with Gamelan Sekar Jaya. Dances performed: Legong Kuntir; Joged.
Apr 2026	UC Berkeley Noon Concert, Berkeley, CA, with the UC Berkeley Gamelan class and Gamelan Sekar Jaya. Dance performed: Pependetan Pengendag.
May 2026	Friends of Indonesia 5th Annual Bazaar, Oakland, CA. Dance performed: Condong.*
May 2026	Gamelan in the Garden, San Francisco, CA, with Gamelan Sekar Jaya. Dances performed: Makepung; Joged Jegog; Kebyar Santi; Legong Karuna Sih.
Jun 2026	Yerba Buena Gardens Festival, San Francisco, CA, with Gamelan Sekar Jaya. Dances performed: Makepung; Kebyar Santi; Joged Jegog.
Jun 2026	Berkeley Exhibition & Festival, Berkeley, CA, with Gamelan Sekar Jaya. Dance performed: Pendet Angklung.

PROFESSIONAL APPOINTMENTS

2026 — Research Associate, Embodied Design Research Laboratory
University of California, Berkeley, USA
Conducted research on embodied mathematics learning, analyzed multimodal data, developed instructional materials and learning designs, contributed to publications and grant-supported projects, and supervised undergraduate researchers.

2021–2025 — Graduate Student Instructor, CalTeach
University of California, Berkeley, United States
Taught and facilitated courses for prospective science and mathematics teachers; supported lesson planning, classroom observation, analysis of student thinking, equity-oriented pedagogy, and teacher-research projects. Contributed to seminar-material preparation, assessment and grading, course evaluation, and curriculum development within CalTeach and in collaboration with CalTeach programs across the University of California system.

2022–2023 — Graduate Research Assistant

XQ Institute, Oakland, California, United States
Supported research on secondary-school project-based learning and school innovation through literature review, qualitative data analysis, and preparation of research materials, publications, and reports.

2019–2021 Assistant Professor, Faculty of Teacher Training and Education
Universitas Mataram, Mataram, NTB, Indonesia
Taught university-level mathematics and mathematics education courses; supervised student projects on developing research in mathematics education; contributed to curriculum development and academic administration; supported academic program and event coordination; and participated in grant development, research projects, international collaborations, and scholarly publications.

2015–2019 Lecturer, Faculty of Mathematics and Natural Sciences
Universitas Pendidikan Ganesha, Singaraja, Bali, Indonesia
Taught university-level mathematics and mathematics education courses; supervised pre-service and in-service teachers on designing inquiry-based mathematics lessons and interactive learning media; contributed to curriculum development and academic administration; supported academic program and event coordination; and co-founded community-based collaborative education projects.

SELECTED PUBLICATIONS

Refereed Journal Articles

Apsari, R. A., & Abrahamson, D. (2026). The emergence of sign–tool hybridity in situated geometry practice: A networking perspective. *Educational Studies in Mathematics, 123*(1), 49–72. https://doi.org/10.1007/s10649-026-10491-8

Apsari, R. A., & Abrahamson, D. (2026, 2024/11/24). Dancing geometry: Imagining auxiliary lines by reflecting on physical movement. *International Journal of Mathematical Education in Science and Technology.* 57(2), 342–369. https://doi.org/10.1080/0020739X.2024.2427099

Abrahamson, D., Lomos, C., Hilkey, S., **Apsari**, **R. A**., & Palatnik, A. (2026). Heteroperception in joint action: Breakdown and reconciliation of collaborating learners' differing views of mathematical objects. *Learning, Culture and Social Interaction, 59,* 101003*.* https://doi.org/10.1016/j.lcsi.2026.101003

Apsari, R. A., Sariyasa, Suharta, I. G. P., Baidowi, & Triutami, T. W. (2021). Pembelajaran penjumlahan dan pengurangan bilangan dua digit secara simultan untuk mengembangkan intuisi bilangan siswa [Simultaneous learning of two-digit addition and subtraction to develop students' number sense]. *Jurnal Tadris Matematika, 4*(1), 31–40. https://doi.org/10.21274/jtm.2021.4.1.31-40

Ndiung, S., Sariyasa, S., Jehadus, E., & **Apsari, R. A.** (2021). The effect of Treffinger Creative learning model with the use of RME principles on creative thinking skill and mathematics learning outcome. *International Journal of Instruction, 14*(2), 873-888.https://doi.org/10.29333/iji.2021.14249a

Apsari, R. A., Sripatmi, Sariyasa, Maulyda, M. A., & Salsabila, N. H. (2020). Pembelajaran matematika dengan media obrolan kelompok multiarah sebagai alternatif kelas jarak jauh [Mathematics learning through multidirectional group chat media as an alternative for distance learning]. *Jurnal Elemen, 6*(2), 318–332. https://doi.org/10.29408/jel.v6i2.2179

Apsari, R.A., Putri, R.I.I., Sariyasa, Abels, M., & Prayitno, S. (2019). Geometry representation to develop algebraic thinking: A recommendation for a pattern investigation in pre-algebra class. *Journal on Mathematics Education, 11*(1), 45-58. http://doi.org/10.22342/jme.11.1.9535.45-58

Journal Articles Under Review/in Preparation

Apsari, **R. A.**, Lomos, C., & Abrahamson, D. (under review). Towards epistemic climate change: Fostering multimodal awareness in educational design.

Apsari, R. A., Kohar, A. W., Pagiling, S. L., Munfarikhatin, A., Assagaf, S. F., Hidayat, D., Hidayat, A. (under review). Rewriting what counts: Transnational teacher educators coordinate funds of knowledge across educational contexts.

Hidayat, A., **Apsari, R. A.,** Kohar, A. W., Pagiling, S. L., Hidayat, D., Assagaf, S. F., Munfarikhatin, A. (under review). Marinated in Western epistemologies: A collaborative autoethnography of Indonesian doctoral students negotiating knowledge across borders

Apsari, **R. A.**, Lomos, C., & Abrahamson, D. (in preparation). Reinventing Thales in Belval: Microgenesis of enactive conjectures in body-scale collaborative geometry activities.

Refereed Conference Proceedings

Apsari, R. A., & Abrahamson, D. (2026). Leveraging heteroperception in collaborative embodied design: The case of body-scale enacting a geometry design [Research report]. In M. Hannula & H. Krzywacki (Chairs). *"Sensing and making sense of mathematics"—Proceedings of the 49th annual meeting of the International Group for the Psychology of Mathematics Education (PME49)*. IGPME.

Apsari, **R. A.**, & Abrahamson, D. (2025). Culturally situated embodied design: A cognitive–anthropological investigation of Balinese spatial practice toward grounding geometry. In R. M. Zbiek, X. Yao, A. McCloskey, & F. Arbaugh (Eds.). *Proceedings of the forty-seventh annual meeting of the North American Chapter of the International Group for the Psychology of Mathematics Education* (pp. 1473–1482). Pennsylvania State University. https://doi.org/10.51272/pmena.47.2025

Apsari, **R. A.**, & Abrahamson, D. (2025). Strings attached: A Balinese dance from attentional anchors to auxiliary constructions. In T. Verhoeff, D. Swart, & S. L. Gould (Eds.), *Proceedings of Bridges Mathematics and the Arts Conference 2025*. Eindhoven University of Technology. http://archive.bridgesmathart.org/2025/bridges2025-539.html

Apsari, **R. A.**, & Abrahamson, D. (2025). Grounding mathematics learning in culturally authentic movement practices: From attentional anchors in Balinese dancing to auxiliary lines in geometry reasoning. In A. Rajala, A. Cortez, R. Hofmann, A. Jornet, H. Lotz-Sisitka, & L. Markauskaite (Eds.), *Proceedings of the 19th meeting of the International Conference of the Learning Sciences (ICLS 2025)*. ISLS.

Apsari, R. A., Sariyasa, Junaidi, Tyaningsih, R. Y., & Gunawan, G. H. (2021). An ethnomathematics case study of Candrasengkala: A reversed-order chronogram in Bali–Java tradition. *Journal of Physics: Conference Series, 1933*, 012075. https://doi.org/10.1088/1742-6596/1933/1/012075

Radiusman, R., Wardani, K. S. K., **Apsari, R. A.**, Nurmawanti, I., & Gunawan, G. (2021). Ethnomathematics in Balinese traditional dance: A study of angles in hand gestures. *Journal of Physics: Conference Series, 1779*, 012074. https://doi.org/10.1088/1742-6596/1779/1/012074

Apsari, R. A., Sariyasa, Indrawan, G., Maulyda, M. A., & Radiusman. (2020). Why should you reverse the order when dividing a fraction? A study of pre-service mathematics teachers' pedagogical content knowledge in fractional concepts. *Journal of Physics: Conference Series, 1503*, 012019. https://doi.org/10.1088/1742-6596/1503/1/012019

Apsari, R. A., Sripatmi, Putri, R. I. I., Hayati, L., & Sariyasa. (2020). From less to more sophisticated solutions: Sociomathematical norms to develop students' self-efficacy. In *Proceedings of the 1st Annual Conference on Education and Social Sciences 2019*, Universitas Mataram. Atlantis Press.

Books and Book Chapters

Parwati, N. N., Suryawan, I. P. P., & **Apsari, R. A.** (2018). *Belajar dan pembelajaran* [Learning and instruction]. *Rajawali Press*.

Apsari, R. A. (2014). Mempertimbangkan kasus ekstrem [Considering extreme cases]. In Y. Hartono (Ed.), *Matematika: Strategi pemecahan masalah* [Mathematics: Problem-solving strategies] (pp. 37–41). Graha Ilmu.

Curriculum Design and Educational Materials

Authored Work

Apsari, **R. A.** (2024). *Menarikan pola bilangan* [Dancing number patterns]. YPMIPA & Penerbit PiBo.

Apsari, **R. A.** (2021). *Buku digital pendidikan matematika realistik Indonesia* [Digital book on Indonesian realistic mathematics education]. YPMIPA. https://bit.ly/RaMEapps

Apsari, R. A. (2020). *Modul belajar numerasi jenjang SD (kelas 4, tema 1–9): Panduan siswa, guru, dan orang tua* [Elementary numeracy learning modules for Grade 4, themes 1–9: Student, teacher, and parent guides]. Pusat Asesmen dan Pembelajaran, Badan Penelitian dan Pengembangan dan Perbukuan, Kementerian Pendidikan dan Kebudayaan [Center for Assessment and Learning, Research and Development and Book Agency, Ministry of Education and Culture]. https://bit.ly/RatihBDR

Apsari, **R. A.** (2020). *Ethnomathematics module for South-East Asian mathematics teachers and pre-service teachers*. SEAMEO QITEP in Mathematics.

Adapted and Translated Work

Gakko Tosho. (2021). *Belajar bersama temanmu: Matematika untuk sekolah dasar kelas IV volume 2—Buku siswa dan buku panduan guru* [Study with your friends: Mathematics for elementary school fourth grade, volume 2—Student book and teacher's guide] (**R. A. Apsari**, Adapt.; I. M. Sulandra, Trans.; M. Isoda, Chief Ed.; 2nd ed.). Pusat Kurikulum dan Perbukuan, Badan Penelitian dan Pengembangan dan Perbukuan, Kementerian Pendidikan, Kebudayaan, Riset, dan Teknologi [Curriculum and Book Center, Ministry of Education, Culture, Research, and Technology].

SELECTED PRESENTATIONS

Invited Conference Presentations

Apsari, **R. A.**, & Abrahamson, D. (2025, Oct.). *Culturally situated embodied design: A framework for grounding mathematical concepts in students' authentic cultural practices*. Invited Speaker. In P. Nilsson (Chair), Design research in mathematics education [Symposium]. Linnæus University, Växjö, Sweden, October 29, 2025.

Apsari, **R. A.** (2025, July). *Culturally situated embodied mathematics design: Tracing the invisible with Geometry Resources in Dance (GRiD)*. Invited Speaker. In I. Kurniasari, B. P. Prawoto, & L. Lestariningsih (Chairs). The 2nd International Conference of Interdisciplinary Research and Studies in Mathematics (Inciresma 2025), Universitas Negeri Surabaya, Indonesia, July 19, 2025

Apsari, **R. A.** (2025, July). *Culturally situated mathematics design: Balinese local spatial practice as a semiotic resource for geometric reasoning*. Invited Speaker. In B. Z. Melani

(Chair), The 7th International Conference on Education and Social Science (ICESS), Universitas Mataram, Indonesia, July 10, 2025

Apsari, R. A. (2024, May). *Geometry-Resources in Dance: Grounding mathematics in Balinese Traditional Dance*. Invited Participatory Presentation. In D. Abrahamson & A. R. Jensenius (Chairs). International Workshop on Embodied Design and Multimodal Learning Analytics, Berkeley, California,

Apsari, R. A. (2023, Nov.). *Reimagine new ways of learning mathematics: Geometry-embodied design situated in the dance tradition*. Invited Speaker. In A. Amrullah (Chair). Seminar Nasional Pendidikan Matematika dan Terapan (National Seminar on Mathematics Education and Application), Universitas Mataram, Indonesia, November 18, 2023.

Refereed Conference Presentations (no proceedings)

Apsari, R. A., & Abrahamson, D. (2026, March). *Flooring dance: From perception to inscription in a culturally situated embodied geometry design*. "Single paper" to be presented at the annual meeting of the European Association for Research on Learning and Instruction (EARLI) Special Interest Groups 10, 17, 21 & 25. University of Augsburg, Augsburg, Germany, March 25–27, 2026.

Apsari, **R. A.** (2024, April). *Grounding geometry as movement discourse: The case of auxiliary constructions in Balinese dance*. In D. Abrahamson (Chair) & S. Gerofsky (Discussant), In-sight out: Challenges and opportunities in learning mathematics through negotiating egocentric and allocentric perspectives. Symposium presented for the SIG Research in Mathematics Education at the annual meeting of the American Educational Research Association, Philadelphia, April 11–14.

Apsari, R. A. (2021, Apr.). *Investigating the potency of Balinese ogoh-ogoh as a context in STEAM education*. Paper presented at the SEAMEO Virtual Congress 2021, Transforming Southeast Asian Education, Science and Culture in the Digital Age, April 28–29, 2021.

Invited Lectures and Seminars

Apsari R. A. (2026, April). *From dance movement to geometric reasoning: Multimodal analysis to trace learning pathways in embodied mathematics design.* Invited lecture at the Department of Mathematics Education, Universitas Pattimura, Maluku, Indonesia, April 28, 2026.

Abrahamson, D., Lomos, C., Hilkey, S., **Apsari**, R. A., & Palatnik, A. (2026, April). Heteroperception in joint action: Breakdown and reconciliation of collaborating learners' differing views of mathematical objects. Invited speaker at SESAME Colloquium, University of California, Berkeley, USA, April 23, 2026.

Apsari, **R. A.** (2026, April). *Getting to know innovative work in educational research: Designing geometry resources in dance (GRiD) learning intervention*. Invited speaker at Cathedral School for Boys, San Francisco: SF's influence on American consciousness—Intersession 2026, Berkeley, USA, April 17, 2026.

Apsari, **R. A.** (2026, April). *Culturally situated embodied mathematics design*. Invited lecture and workshop facilitator at Knowing and Learning in Mathematics and Science Undergraduate Course, University of California, Berkeley, USA, April 8, 2026.

Apsari R. A. (2026, Mar.). *Conducting interdisciplinary mathematics education research: Insights from a culturally situated embodied geometry design.* Invited presentation at Workshop on Scientific Writing: From Creative Research to Publication, Graduate School of Mathematics Education, Universitas Sebelas Maret, Solo, Indonesia, March 27, 2026.

Apsari, **R. A.** (2025, Dec.). *Research on grounding mathematics in cultural practice: The case of GRiD dancing-geometry project*. Invited presentation at Webinar Series in Collaborative Research in Mathematics Education, organized by the Indonesian Group for Research and Learning in Mathematics Education (IGRiME) and the Mathematics Department of Universitas Negeri Makassar (UNM), Indonesia, December 15, 2025.

Apsari, **R. A.** (2025, Dec.). *From tradition to transformation: Embodied cultural design for the primary school mathematics classroom*. Invited lecture at Primary Education Department, BINUS University, Jakarta, Indonesia, December 9, 2025

Apsari, **R. A.** (2025, Mar.). *Culturally situated embodied mathematical design*. Invited lecture at Knowing and Learning in Mathematics and Science Undergraduate Course, University of California, Berkeley, USA, March 12, 2025.

Apsari, **R. A.** (2024, Dec.). *Dancing geometry: Embodied design for culturally situated mathematics classroom*. Invited lecture at Elementary School Teacher Education Department, Universitas Islam Raden Rahmat, Malang, Indonesia, December 13, 2024.

Apsari, **R. A.** (2024, July). *GRiD: Geometry resources in dance*. Invited presentation at Copernicus Science Center, Warsaw, Poland, July 23, 2024.

Apsari, R. A. (2023, Dec.). *Pembelajaran berdiferensiasi dalam Kurikulum Merdeka* [Differentiated instruction in the Indonesian National Curriculum]. Invited presentation at the National Webinar on Mathematics Education, organized by the Mathematics Education Study Programs of Universitas Hasyim Asy'ari and Universitas Mataram, Indonesia, December 2, 2023.

Apsari, **R. A.** (2023, Nov.). *Dancing mathematics*. Invited presentation at Scientific Adventures for Girls, Washington Elementary School, Berkeley, USA, November 16, 2023.

Apsari, R. A. (2023, Oct.). *Penyusunan asesmen dalam Kurikulum Merdeka* [Designing assessment in the Indonesian National Curriculum]. Invited presentation at the Regional Academic Seminar for Senior High School Mathematics Teachers, SMAN 1 Singaraja, Indonesia, October 11, 2023.

Apsari, R. A. (2023, Jan.). *Merancang kelas berbasis proyek untuk penguatan Profil Pelajar Pancasila* [Designing project-based classrooms to strengthen the Pancasila Student Profile]. Invited presentation at the National Online Training Program for Indonesian Teachers, January 26–27, 2023.

Apsari, R. A. (2022, Apr.). *Integrating history of science, technology, and society into the mathematics classroom*. Invited presentation at the Doctoral Mathematics Education Study Program, Universitas Pendidikan Indonesia, April 23, 2022.

Apsari, R. A. (2022, Mar.). *Belajar dari sejarah: Merancang kelas berbasis proyek untuk penguatan profil Pelajar Pancasila* [Learning from history: Designing project-based classrooms to strengthen the Pancasila Student Profile]. Invited presentation at the Workshop Nasional Pembelajaran Paradigma Baru [National Workshop on New Paradigm Learning], Mathematics Education Study Program, Universitas PGRI Mahadewa Indonesia, March 28, 2022.

Apsari, R. A. (2020, July). *Fenomena didaktis pada PMRI* [Didactical phenomenology in Indonesian Realistic Mathematics Education]. Invited presentation at the Seminar Nasional Meninjau Kembali PMRI [National Webinar on Revisiting RME], organized by the Alumni Association and the Department of Mathematics, Universitas Pendidikan Ganesha, Indonesia, July 2, 2020.

Refereed Poster Presentation

Apsari, **R. A.** (2025). *Geometry Resources in Dance (GRiD): Embodied design-based research to foster the construction of auxiliary lines in solving geometry problems from tacit attentional anchors in movement practice*. The International Conference of the Learning Sciences. ISLS.

Apsari, R. A. (2024). *Grounding auxiliary geometrical constructions as semiotic articulation of tacit attentional anchors: The case of Balinese dance.* Research Day, Berkeley School of Education, University of California, Berkeley.

Refereed Conference Workshops

Apsari, **R. A.** (2025, July). *Dancing geometry*. Facilitator in K. Schaffer (Program Coordinator). Dancing and Mathematics Family Day Activity, Bridges: Mathematics and Art Conference [Workshop], Eindhoven, the Netherlands, July 17, 2025.

Invited Workshops

Apsari, **R. A.** (2025, July). *Culturally situated embodied mathematical design*. In Skrzypowska, J. (Organizer). Workshop for Educators and Researchers at the Tinkatorium Station, Copernicus Science Center, Warsaw, Poland, July 4, 2025

Apsari, **R. A.**, Surjorahardjo, I. J., & Wirjadi, K. (2025, April). *Dancing geometry*. In K. Graf & L. Jacoby (Co-chairs). Expand Your Horizon at Berkeley, Workshop for 5th–8th grade girls, trans youth, and gender non-conforming and/or non-binary students, University of California, Berkeley, USA, April 12, 2025

Apsari, **R. A.** (2024, May). *Balinese dance and embodied mathematical design* [Participatory presentation]. In D. Abrahamson & A. R. Jensenius (Workshop Chairs), Embodied Design and Multimodal Learning Analytics, University of California, Berkeley, USA, May 9, 2024.

Apsari, R. A. (2023, Oct.). *STEM in your classroom: NetLogo and agent-based modeling for real-world phenomena*. Invited workshop at Pendekatan STEM/STEAM untuk Pembelajaran Matematika [STEM/STEAM approaches to mathematics learning], Master's Program in Mathematics Education, Universitas Pendidikan Ganesha, October 15, 2023.

SELECTED PROJECTS & GRANTS

Funded Research Projects

Geometry Resources in Dance (GRiD)
Principal researcher; advisor: Professor Dor Abrahamson
Developed a low-cost gridded-floor learning environment to transform tacit attentional anchors in movement coordination into auxiliary lines for geometric reasoning.
Funding: 2026, Dissertation Grant, Indonesia Endowment Fund for Education ($6,000)

Digital GRiD: Real-Time Movement Tracing for Embodied Geometry Learning
Principal investigator; collaborator: Yangyang Yang, University of California, Berkeley
Developed a digital movement-tracing system to provide real-time visual feedback during embodied geometry activity.
Funding: 2026, ICBS Interdisciplinary Grant, Institute of Cognitive and Brain Sciences, University of California, Berkeley, USA ($4,000); 2024, Innovation Catalyst Grant, Jacobs Institute for Design Innovation, University of California, Berkeley, USA ($500)
GRiD with Real-Time Mobile Eye-Tracking

Principal investigator; collaborator: Dr. Catalina Lomos (Luxembourg Institute of Socio-Economic Research), Assistant Professor Alik Palatnik (The Hebrew University of Jerusalem), Professor Dor Abrahamson
Designed and investigated collaborative embodied geometry learning through multimodal interaction analysis and mobile eye tracking.
Funding: 2025, Teacher education for embodied cognition in mathematics—a multi-modal data digitally enhanced micro-process study, Luxembourg National Research Fund (FNR), Luxembourg (facilities and material resources)

Collaborative Self-funded Research Project
Dance-mathematics phenomenological study (2025–present)
Collaborative researcher with Dr. Anna Shvarts (Utrecht University), Dr. Shabari Rao (National Institute of Advanced Studies), Dr. Janani Suresh Ram (Germany), Francesca Neri Macchiaverna (Università Roma Tre), Dr. Eka Zharinova (United Kingdom)
Examines how dancers experience, articulate, and coordinate mathematical and aesthetic relations through movement.

Dance as a Resource in Mathematics Education (2026–present)
Project lead; collaborators: Dafna Efron (Hebrew University of Jerusalem), Dr. Shabari Rao, (National Institute of Advanced Studies), and Shanti Pise (University of Hertfordshire)
Leads an international collaborative inquiry into how dance practices can function as epistemic and pedagogical resources in mathematics education.

Funds of Knowledge of Transnational Mathematics Teacher Educators (2026–present)
Project lead; Collaborators: Dr. Angga Hidayat (Florida Gulf Coast University), Ahmad Wachidul Kohar (Michigan State University), Anis Munfarikhatin (Syracuse University), Dayat Hidayat (Purdue University), Said Fachry Assagaf (Indiana University Bloomington), and Sadrack Pagiling (Michigan State University)
Uses collaborative autoethnography and iterative narrative-reflective dialogue to examine how transnational mathematics teacher educators reconstruct professional knowledge across educational contexts.

Institutional Development Grants
Development of the Merdeka Belajar–Kampus Merdeka Curriculum for the Mathematics Education Program, Universitas Mataram, Indonesia
Core project team member
Led project development and reporting from proposal preparation through the interim and final reports, while contributing to curriculum mapping, credit-equivalency structures, mobility and school-internship pathways, implementation procedures, and partnership agreements.
Funding: 2020, Directorate of Learning and Student Affairs, Indonesian Ministry of Education and Culture, IDR 60,000,000.

Lesson Study for Learning Community with Secondary Mathematics Teachers, Mataram, Indonesia
Project team member and teacher-development facilitator
Coordinated and facilitated two cycles of collaborative lesson planning, enactment, observation, reflection, and redesign with mathematics teachers in the region of

Sandubaya–Cakranegara professional community. Also supported training in digital tools for online mathematics instruction during the COVID-19 pandemic.
Funding: 2020, Universitas Mataram, IDR 5,000,000.

Development of Online Learning Systems and Course Materials for Linear Algebra and Basic Mathematics, Universitas Mataram, Indonesia
Course development team member
Co-developed university-level online courses in Linear Algebra and Basic Mathematics through Universitas Mataram's SPADA learning-management system, including course structure, digital instructional materials, learning activities, and online delivery.
Funding: 2020, SPADA Online Learning System Grant, Universitas Mataram.
Evaluation of the Non-profit Taman Cerdas Ganesha Program, Bali, Indonesia
Project coordinator
Prepared the grant proposal and coordinated the implementation of a departmental study evaluating the educational effects of Taman Cerdas Ganesha, a community-based STEM initiative I had co-founded and led since 2017. Coordinated data collection and project activities with participating students, volunteers, local schools, and village partners.
Funding: 2018, Universitas Pendidikan Ganesha, IDR 10,000,000.

Training and Mentoring in the Design and Implementation of Interactive Mathematics Instruction for Prospective Mathematics Teachers, Bali, Indonesia
Project coordinator
Prepared the grant proposal and coordinated a department-supported community-engagement program that trained prospective mathematics teachers to design and facilitate interactive mathematics learning activities. Organized training, instructional-material development, volunteer preparation, and supported implementation through Taman Cerdas Ganesha and its partner communities.
Funding: 2018, Universitas Pendidikan Ganesha, IDR 7,500,000.

Lesson Study for English for Mathematics Course, Universitas Pendidikan Ganesha, Indonesia
Project team member and assistant to the model lecturer
Contributed to the design and implementation of a department-level lesson study project integrating the Initiation–Construction and Reconstruction–Creation–Presentation–Reflection (ICRCPR) instructional model, blended learning, and portfolio assessment. Participated in three iterative cycles of collaborative research-lesson planning, materials development, classroom enactment, observation, documentation, reflection, and redesign.
Funding: 2016, Universitas Pendidikan Ganesha, IDR 5,000,000.

TEACHING EXPERIENCES

University of California, Berkeley, *Graduate Student Instructor*
- History of Science (Fall 2021)
- History of Science, Technology, and Society (Spring 2022)
- Project-Based Instruction (Fall 2022)
- Knowing and Learning in Science and Mathematics (Spring 2023, Fall 2023)
- Classroom Interactions in Science and Mathematics (Spring 2024–Spring 2025)

Universitas Mataram, *Assistant Professor*
- Calculus II (Fall 2019)

- Elementary Mathematics (Fall 2019, Fall 2020)
- Linear Algebra (Fall 2019, Fall 2020)
- Elementary Statistics (Summer 2019, Summer 2020, Spring 2021)
- Abstract Algebra (Spring 2020)
- English for Mathematics (Spring 2020, Spring 2021)
- Educational Technology in Mathematics Education (Spring 2020, Spring 2021)
- Mathematical Learning and Instruction (Fall 2020)
- Calculus I (Summer 2021)
- Research Methodology (Spring 2021)

Universitas Pendidikan Ganesha, *Lecturer*

- Complex Analysis (Fall 2015, Spring 2017, Spring 2018)
- English for Mathematics (Spring 2016, Fall 2016, Spring 2017, Spring 2019)
- Discrete Mathematics (Spring 2016)
- Foundations of Education (Fall 2016, Spring 2017, Spring 2019)
- Learning and Instruction (Fall 2016, Fall 2017)
- Entrepreneurship in the Educational Sector (Spring 2017, Spring 2019)
- Curriculum and Instruction (Fall 2018)

FELLOWSHIPS & PROFESSIONAL DEVELOPMENT FUNDING

Fellowship

2021–2026 Doctoral Fellowship, Indonesia Endowment Fund for Education (Lembaga Pengelola Dana Pendidikan), Full Ph.D. funding.

2013–2015 StuNed & DIKTI Awardee, StuNed Scholarship Programme & Directorate General of Higher Education (DIKTI), Indonesia, Full Master's funding

2008–2012 PPA DIKTI Awardee, Directorate General of Higher Education (DIKTI), Indonesia, Partial Bachelor funding.

Professional Development Funding

2026 JURE Research Development Fund, The European Association for Research on Learning and Instruction (EARLI) to attend PME 49 Conference (€1,000)

2026 Student Opportunity Fund, University of California, Berkeley, to attend EARLI SIG Conference ($2,000)

2025 Graduate Assembly, University of California, Berkeley, to attend ISLS, PME-NA, and Linnæus University Symposium ($4,000)

2025 Student Travel Scholarship, Bridges: Mathematics and Art Conference ($800)

2024 Student Opportunity Fund to attend AERA Conference ($1,000)

SERVICES

Peer Review for International Journals

- Instructional Science
- Humanities and Social Sciences Communications
- Mathematics Education Research Journal
- International Journal of Mathematical Education in Science and Technology
- Perceptual and Motor Skills
- Journal of Mathematics Education
- Southeast Asian Mathematics Education Journal
- Journal of Research and Advances in Mathematics Education

- International Journal of Instruction
- Mathematics Education Journal

Peer Review for International Conference

- American Educational Research Association (AERA) Annual Meeting 2027 (SIG-Research in Mathematics Education and Division Postsecondary Education)
- North American Chapter of the International Group for the Psychology of Mathematics Education (PME-NA) 2027
- International Conference on Learning Sciences (ICLS) by the International Society of Learning Sciences (ISLS) 2026
- European Society for Research in Mathematics Education (ERME) Topic Conference 19: Mathematics Teaching and Teacher Practice(s) 2026
- European Association for Research on Learning and Instruction (EARLI) SIG 10, 17, 21, and 25 Conference 2026
- International Conference on Mathematics and Learning Research (ICOMER), Universitas Muhammadiyah Surakarta, 2022
- International Conference on Science Education and Sciences (ICSES), Universitas Mataram, 2021
- International Conference on Mathematics and Natural Sciences (IConMNS), Universitas Pendidikan Ganesha, 2019

Peer Review for Indonesian Accredited National Journals

- Inomatika (Inovasi Matematika), STKIP Muhammadiyah Bangka Beltung
- JPM Rafa, UIN Raden Fatah Palembang
- Mandalika, Universitas Mataram
- AlphaMath: Journal of Mathematics Education, Universitas Muhammadiyah Purwokerto
- JUPITEK (Jurnal Pendidikan Matematika), Universitas Pattimura
- BETA Jurnal Tadris Matematika, UIN Mataram
- EDUMATICA, Universitas Jambi
- Jurnal Pendidikan dan Pembelajaran Matematika Indonesia (JPPMI), Universitas Pendidikan Ganesha
- Mosharafa: Jurnal Pendidikan Matematika, Institut Pendidikan Indonesia
- Cartesian: Jurnal Pendidikan Matematika, Universitas Hasyim Asy'ari
- PLUSMINUS: Jurnal Pendidikan Matematika, Institute Pendidikan Indonesia
- Indiktika Jurnal Pendidikan Matematika, Universitas PGRI Palembang
- Delta-Pi: Jurnal Matematika dan Pendidikan Matematika, Universitas Khairun
- International Journal of Review in Mathematics Education

Editor for Indonesian Accredited National Journals

- BETA Jurnal Tadris Matematika, UIN Mataram

Undergraduate Students Research Mentoring

Mentored 23 UC Berkeley students through research-methods courses and research apprenticeship programs, and provided additional mentoring to students from other institutions who independently contacted the Embodied Design Research Laboratory. Areas of mentoring included qualitative research, multimodal analysis, educational design, embodied mathematics learning, and research communication.

- Research Methods for Science and Mathematics K–12 Teachers, UC Berkeley
  Fall 2023, Spring 2024, Fall 2025, Spring 2026 — 13 students

- Undergraduate Research Apprentice Program, UC Berkeley
  Fall 2024–Spring 2026 — 2 students
- Embodied Design Research Laboratory
  Fall 2023–Summer 2026 — 8 students from Universitas Airlangga, Carleton College, De Anza College, and UC Berkeley

International Conferences Organization

2026	Session Chair, PME 49
2025	Hybrid Ambassador, International Conference of the Learning Sciences (ICLS), ISLS
2020	Secretary, Annual Conference on Education and Social Sciences (ACcESS), Universitas Mataram
2017	Secretary, International Conference on Mathematics and Natural Sciences (IConMNS), Universitas Pendidikan Ganesha

Selected Community Services

Indonesian Group for Research and Learning in Mathematics Education (IGRiME)
Head of Communications Division (2024–present)
Lead IGRiME's science-communication strategy, translating mathematics-education research into accessible public-facing content for researchers, teachers, students, and wider audiences; support members in communicating their expertise and oversee editorial planning, content development, and digital dissemination across the organization's social-media platforms.

KASI PANDU, Indonesia Diaspora Students Association (PPI Dunia)
Mentor (2024)
Designed and facilitated accessible STEM learning activities for elementary-school students, combining instructional-material development with direct mentoring.

Indonesia Mengglobal
Digital Communications Manager (2017–2020); Director of Digital Communications (2020–2023); Contributor and Mentor (2023–present)
Led and contributed to the curation, translation, and dissemination of accessible educational content for Indonesians pursuing international study and professional opportunities. Helped develop the inaugural PhD Preparation Program curriculum in 2020 and subsequently supported the organization as a writer, speaker, assessor, and mentor in its master's and PhD preparation programs.

Taman Cerdas Ganesha
Co-Founder and Project Lead (2017–2019)
Co-founded and led a community-based initiative providing free STEM learning opportunities for elementary-school students in rural Bali while creating opportunities for pre-service teachers to design materials, teach, and reflect on their pedagogical practice. Directed project planning and implementation, developed partnerships with neighboring schools and village communities, recruited and trained volunteers, cultivated donor relationships, and prepared grant proposals to support program development and sustainability.

LANGUAGES

Indonesian — Native proficiency
Balinese — Conversational proficiency
English — Full professional proficiency
German — Basic proficiency

PROFESSIONAL MEMBERSHIP

- American Educational Research Association (AERA)
  Special Interest Group Research in Mathematics Education (RME)
- European Association for Research on Learning and Instruction (EARLI)
  Special Interest Group 10 (Social Interaction in Learning and Instruction) and SIG 25 (Educational Theory)
- International Group for the Psychology of Mathematics Education (PME)
- North American Chapter of the International Group for the Psychology of Mathematics Education (PME-NA)
- Indonesian Mathematical Society (IndoMS)
- Indonesian Mathematics Educators Society (I-MES)
- Indonesian Group for Research and Learning in Mathematics Education (IGRiME)

MEDIA HIGHLIGHTS

2026 Berkeley School of Education (YouTube, BSE in Focus, Ratih Ayu Apsari)
https://youtu.be/ptWgkw0Necs?si=g0KPIt1nU_Kf3BYP

REFERENCES

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Director of CalTeach
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emstone@berkeley.edu

Ami Shulman [Dance Reference]
Guest teacher at the Juilliard School
Certified senior Feldenkrais® Practitioner
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